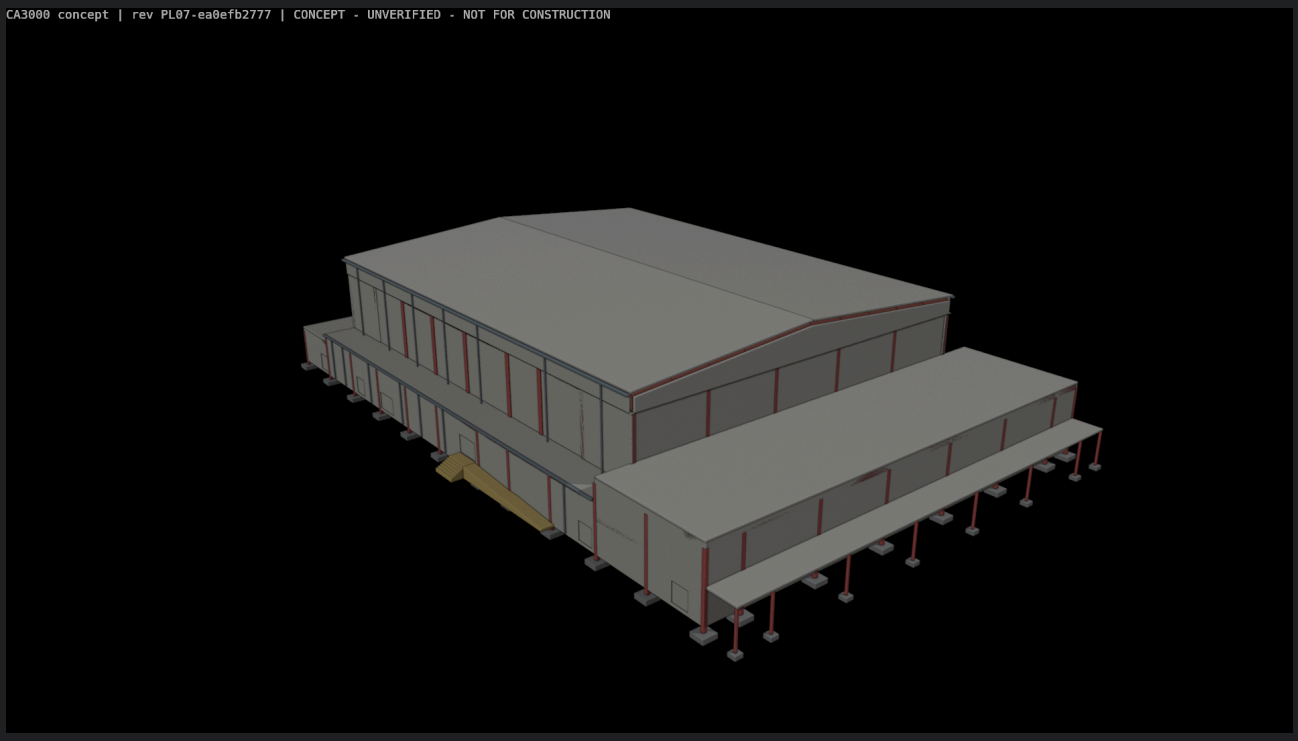
PRELIMINARY ARCHITECTURAL & STRUCTURAL ENGINEERING DESIGN PROPOSAL

Proposed 3,000-seat church auditorium Option A — end-stage auditorium



DESIGN ID A-END-STAGE · MODEL REVISION PL08-590D273E35 · ZIMBABWE (SITE NOT IDENTIFIED) · 2026-11-02

CONCEPT — UNVERIFIED · NOT FOR CONSTRUCTION. Reviewer scores are QA metrics, not certification.

Document control & provenance

Design ID	A-end-stage
Model revision	PL08-590d273e35
Manifest SHA-256	155c6b8dbb09bdd4c70fd9c34f65417aca5e06663b3a7d49e3848c3f509ca95c
Blender	5.2.3 LTS
Units	metres
Status	CONCEPT - UNVERIFIED - NOT FOR CONSTRUCTION
Build	scripts/run_pipeline.py --iteration N --tag PL (writes only inside this folder; Blender from BLENDER_EXE)
Measurement provenance	Every figure, table and drawing in this document is generated from this design's Blender manifest, its verified reports (geometry_checks, blend_verification, ifc_validation) and comparisons/compare.py run on the same manifest.

Source files hashed in the revision record

Source	SHA-256 (16)
config/auditorium_params.json	ae79bb7a6bea4397
scripts/auditorium_layout.py	d9306ca6bcfbcdd3
scripts/build_auditorium.py	988f3187f2ccd364
scripts/verify_blend.py	5d62c1b9ab72485c
scripts/export_ifc.py	f568e61847fb12c3
scripts/validate_ifc.py	a77ebccf63bd47e2
scripts/make_drawings.py	82dd1dabb80004fa
scripts/generate_reports.py	31c1a3f6a814f2e2
scripts/run_pipeline.py	44c1907d15e9911a
tests/test_layout.py	2d6c6de4700f2a8a

Concept and scope

The platform stands at one end of a 50 m wide hall; 41 straight rows rise to an upper foyer at the rear. 9 steel trusses of 50.0 m span the hall width.

Approved scope changes (client decisions, recorded in DESIGN_BRIEF.md)

- 2026-11-01 - Choir location (post-loop revision PL01): the choir is placed on the two sides of the platform, facing inward across it, not at the rear of the stage. The rear of the platform is kept free for the band / musicians. Choir seats remain excluded from the 3,000 congregation seats. Choir size (144) remains a concept assumption pending client confirmation.
- 2026-11-01 - Choir seating removed (post-loop revision PL03): the fixed choir chairs and side choir risers are removed from the platform. The whole 30 m platform is one open level for the choir (loose chairs or standing) and the band / musicians, reached step-free by the platform ramp. The platform occupant allowance (200, for egress) is unchanged. This supersedes the PL01 side-choir layout.
- 2026-11-01 - Choir in the platform wings (post-loop revision PL04): 144 choir chairs (72 per side, 6 rows x 12) are placed in the two platform wings on the hall floor, beside the stage, facing inward towards the platform. The platform itself stays one open level for the band / musicians. The choir is step-free with one wheelchair choir place per wing, and each block keeps at least 1.2 m clear of the platform, its stair / ramp, walls, columns and doors. Choir seats remain excluded from the 3,000. This supersedes PL03 (no choir seating) and PL01 (choir risers on the platform sides).
- 2026-11-01 - Smaller platform (post-loop revision PL05): the platform is reduced to 20 m wide x 10 m deep (was 30 m x 15.6 m), keeping the same front edge, centre line, height (+0.90 m) and pulpit position, so the 3,000 seats and their sightlines are unchanged. The wing choir blocks move forward to sit alongside the platform. The floor freed behind and beside the platform is left empty for now - no extra chairs.
- 2026-11-02 - Larger choir round the platform (post-loop revision PL06): following the client's mark-up, the choir fills four zones round the platform sides - on each side, a lower block (between the front cross passage and a 1.5 m cross aisle) and an upper block (from the cross aisle back to a rear strip). Chairs are angled towards the pulpit. Exit routes stay clear: a 2.4 m corridor along each hall wall to the front, wing and backstage doors, the front cross passage, the cross aisle and the rear strip, with 1.2 m round the platform, its stair and ramp. The six front wheelchair places are kept and there is one wheelchair choir place each side. Choir occupants now count in the egress tally (choir chairs + wheelchair choir places + 60 platform party).
- 2026-11-02 - Platform against the rear wall (post-loop revision PL07): the 20 m x 10 m platform is pushed back to touch the stage rear wall; the pulpit moves with it. The congregation seating stays where it is, leaving 5.85 m of open floor in front of the platform. Because the sightline target (the platform front edge) is now further from the front rows, only row 1 stays level and the platform is raised to 1.00 m (from 0.90 m), which keeps the rest of the rake within the existing ramps and stairs. The platform ramp now lands at the rear east corner and runs down towards the front; the platform stair is at the rear west corner; both side edges of the platform are guarded, and the front edge is left open as the preaching / performance edge (to be risk-assessed).
- 2026-11-02 - Stair, ramp and choir rows (post-loop revision PL08): the platform stair moves to the west side at the front corner of the platform. The accessible ramp runs along the stage rear wall, from the platform's east edge towards the east wall (the platform is its top landing); the east backstage door moves just east of the ramp foot. Choir chairs stand in straight rows facing across to the platform (no angle). The two lower choir blocks stay between the cross aisle and the front passage (exits kept clear) and are trapezia, their platform-side edge slanting back 2 m towards the front passage. Wheelchair choir places move to the wall end of the cross aisle.

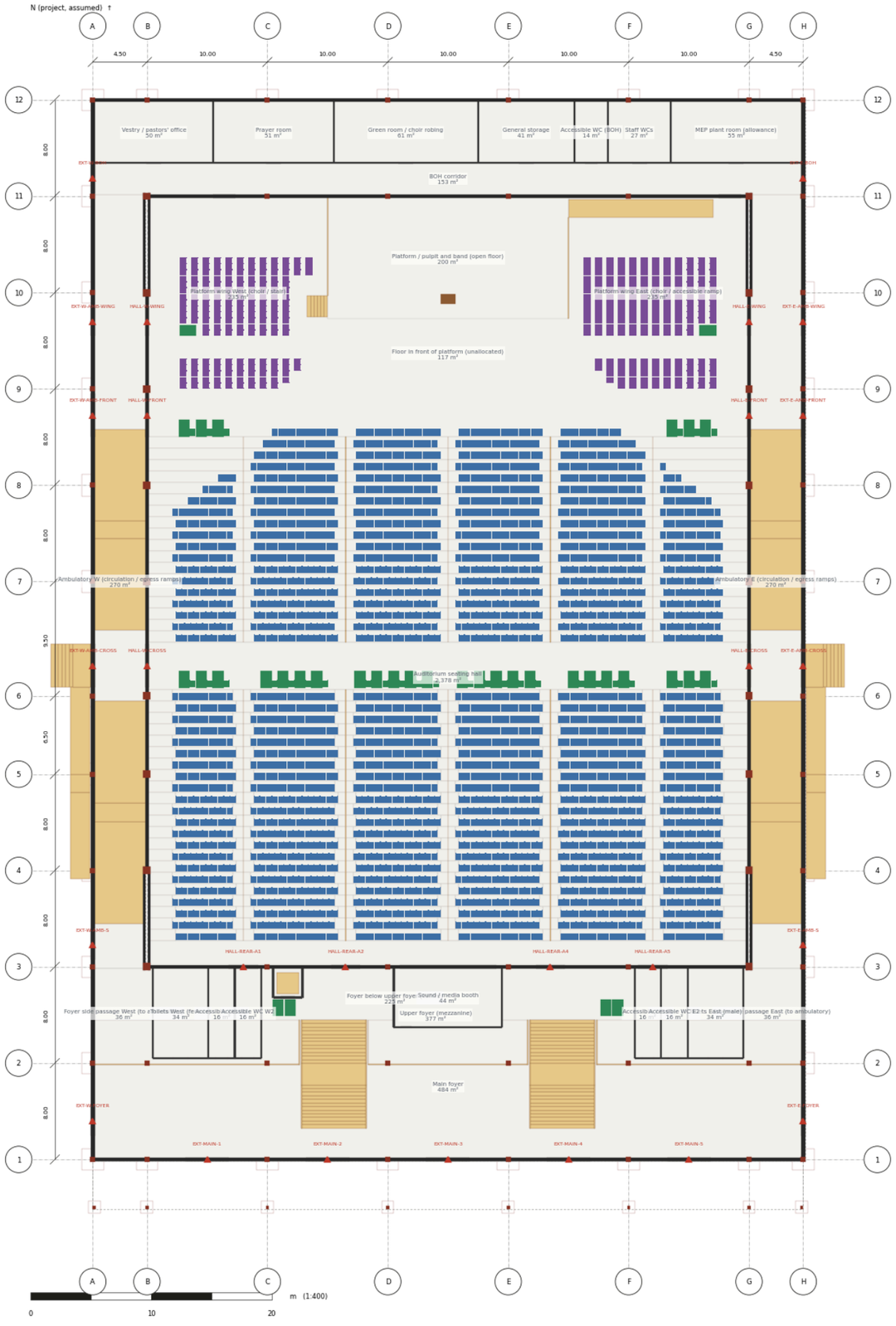
Project at a glance

Measured by comparisons/compare.py from this design's manifest (identical code for both options).

Measure	Value
Congregation seats (verified: layout, .blend, IFC, CSV)	3000
Rows	41
Eye-to-pulpit distance mean / median / max (m)	35.52 / 36.83 / 57.58
Seats within 20 m / 30 m of the pulpit	13% / 38%
Head turn to pulpit mean / max (deg)	20.5 / 52.5
Max bearing from the pulpit (deg)	52.5
Min C-value n1 / n2 (from chair objects)	0.0603 / 0.1206
Footprint (m)	59.3 x 92.3
Hall floor area / per seat (m2)	3165.9 / 1.055
Hall air volume / per seat (m3)	49768 / 16.59
Roof area (sloped, m2) / top (m)	5332.3 / 18.4
Trusses / span (m)	9 / 50.0
Columns / pad footing concrete (m3)	76 / 163.4
Hall exits / door width (m)	10 / 28.8
Wheelchair spaces / lifts	30 / 1
Choir chairs / platform area (m2)	386 / 200.0

Verification

Source	Result
Layout engine checks	{"PASS": 37, "FAIL": 0, "WARN": 4}
Re-opened .blend checks	{"PASS": 18, "FAIL": 0, "WARN": 0}
IFC validation	{"PASS": 21, "FAIL": 0, "WARN": 0}



PROJECT

3,000-Seat Church Auditorium - Coordination Concept

DESIGN

A-end-stage

MODEL REVISION

PL08-590d273e35

DRAWING

Ground floor plan

Plan at L00, cut below the roof structure

SCALE

1:400 at A3

SHEET

A-P100

SOURCE

`blender_manifest.json sha256 155c6b8dbb09bdd4`

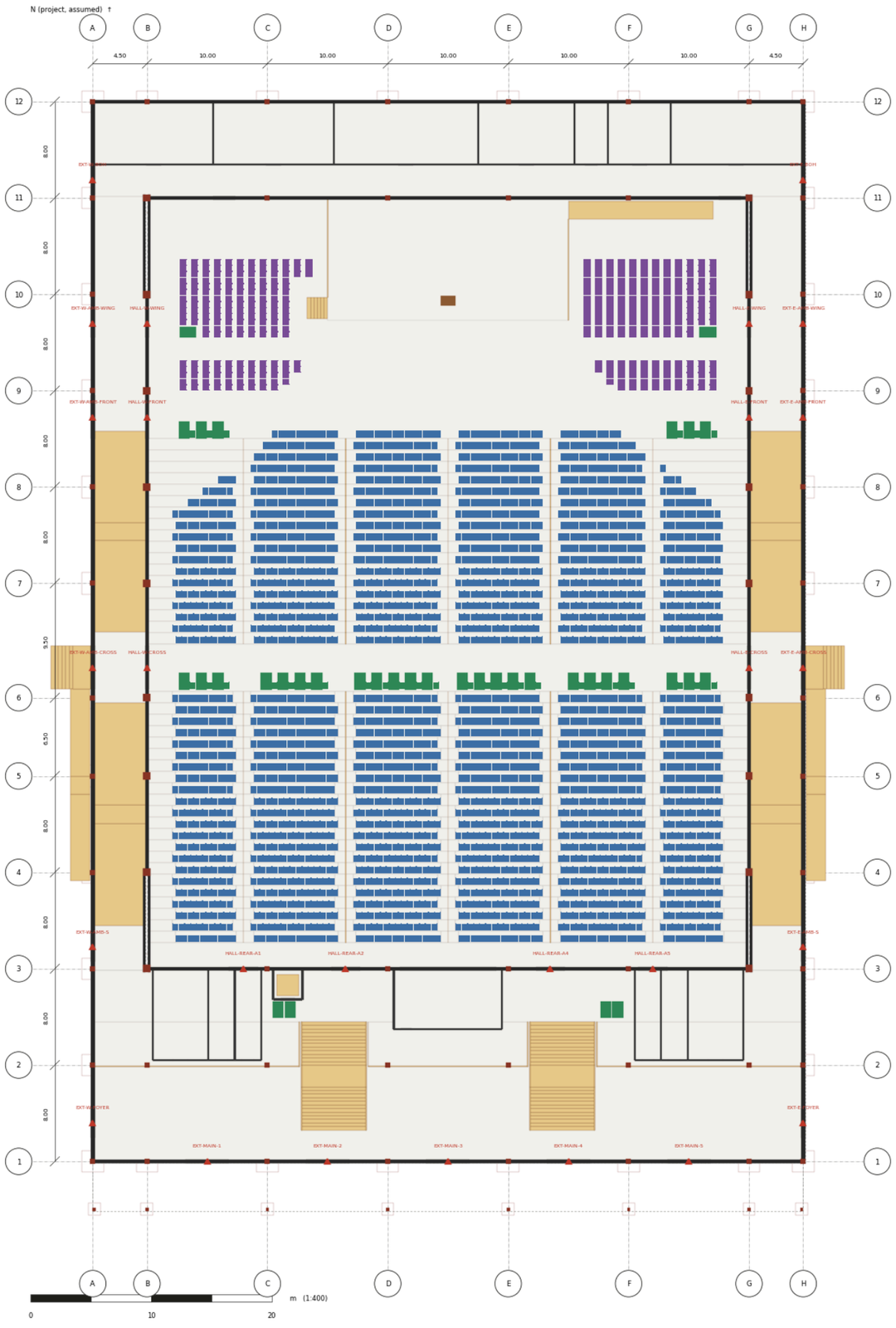
UNITS

metres; levels in m above L00

STATUS

CONCEPT - UNVERIFIED - NOT FOR CONSTRUCTION

Concept coordination drawing generated from the Blender model. Not for construction. Dimensions unverified pending professional review.



PROJECT

3,000-Seat Church Auditorium - Coordination Concept

DESIGN

A-end-stage

MODEL REVISION

PL08-590d273e35

DRAWING

Seating, aisles and exits

Plan at L00, cut below the roof structure

SCALE

1:400 at A3

SHEET

A-P110

SOURCE

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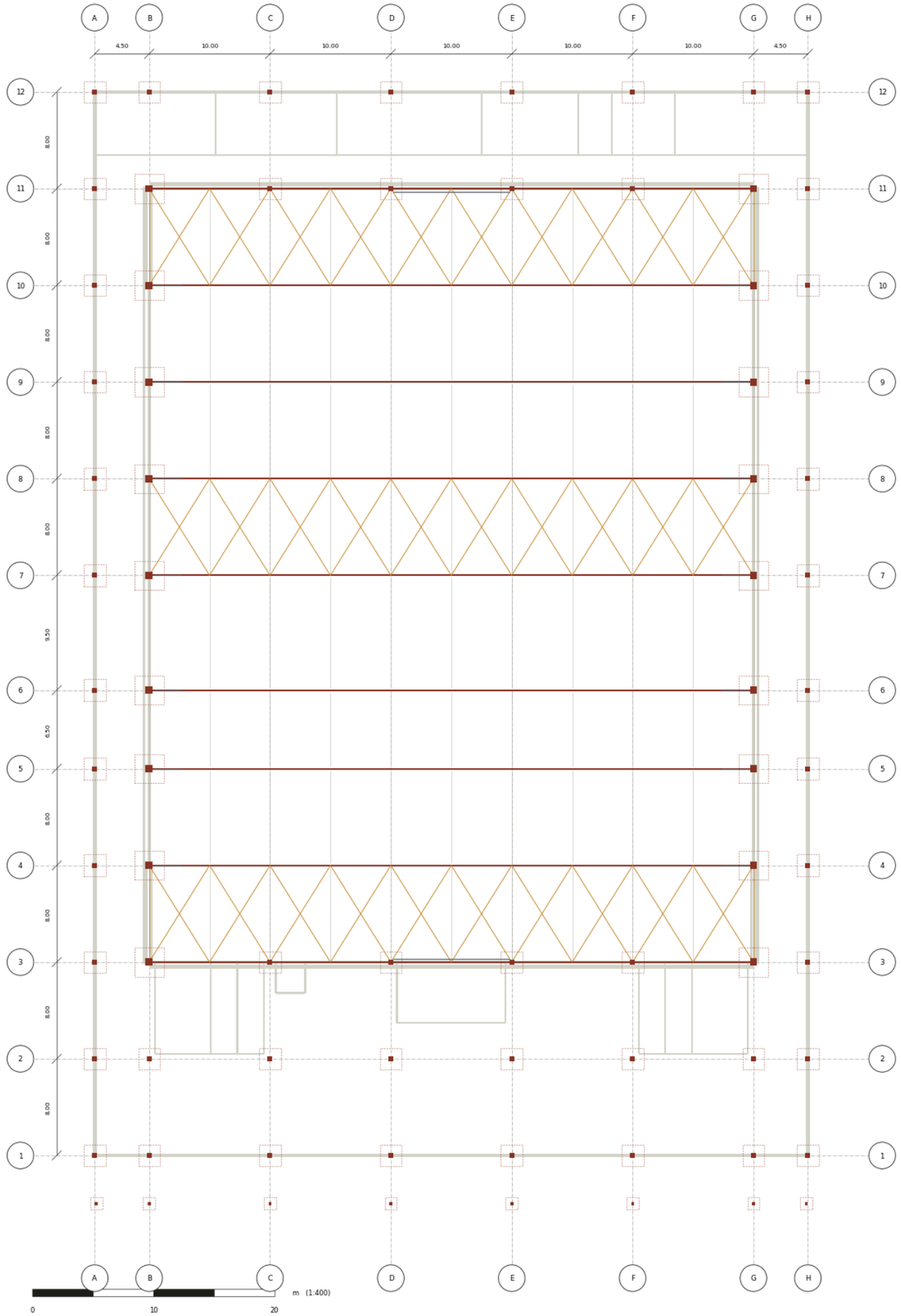
UNITS

metres; levels in m above L00

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PROJECT

3,000-Seat Church Auditorium - Coordination Concept

DESIGN

A-end-stage

MODEL REVISION

PL08-590d273e35

DRAWING

Structural grid and roof framing

Columns, pad footings (dashed), trusses, purlins and bracing

SCALE

1:400 at A3

SHEET

A-S100

SOURCE

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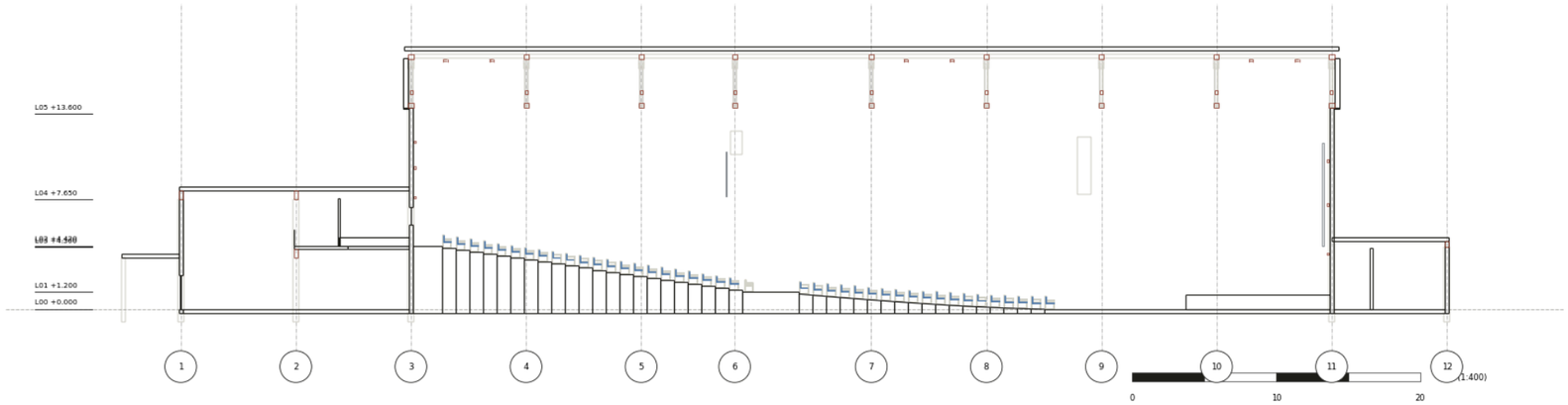
UNITS

metres; levels in m above L00

STATUS

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Concept coordination drawing generated from the Blender model. Not for construction. Dimensions unverified pending professional review.



PROJECT

3,000-Seat Church Auditorium - Coordination Concept

DESIGN

A-end-stage

MODEL REVISION

PL08-590d273e35

DRAWING

Long section

Cut on x = 31.00 m, looking towards +x

SCALE

1:400 at A3

SHEET

A-S300

SOURCE

blender_manifest.json sha256 155c6b8dbb09bdd4

UNITS

metres; levels in m above L00

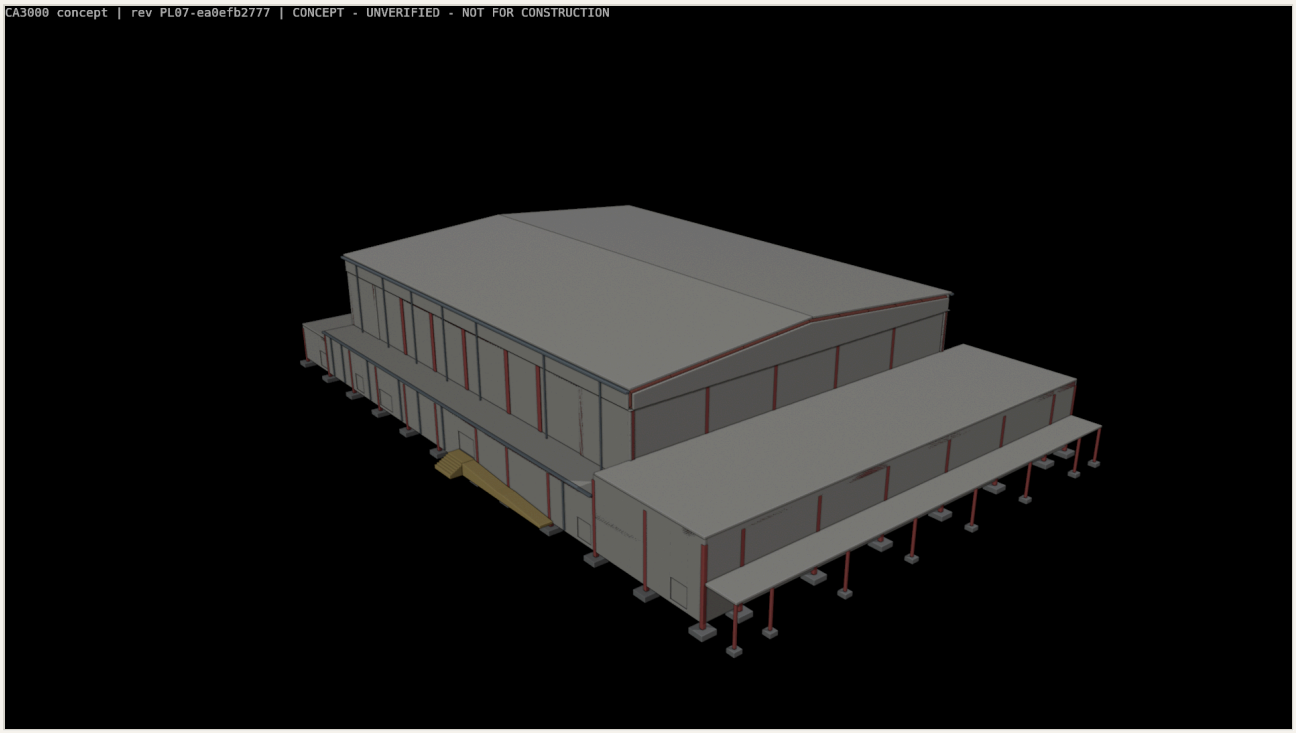
STATUS

CONCEPT - UNVERIFIED - NOT FOR CONSTRUCTION

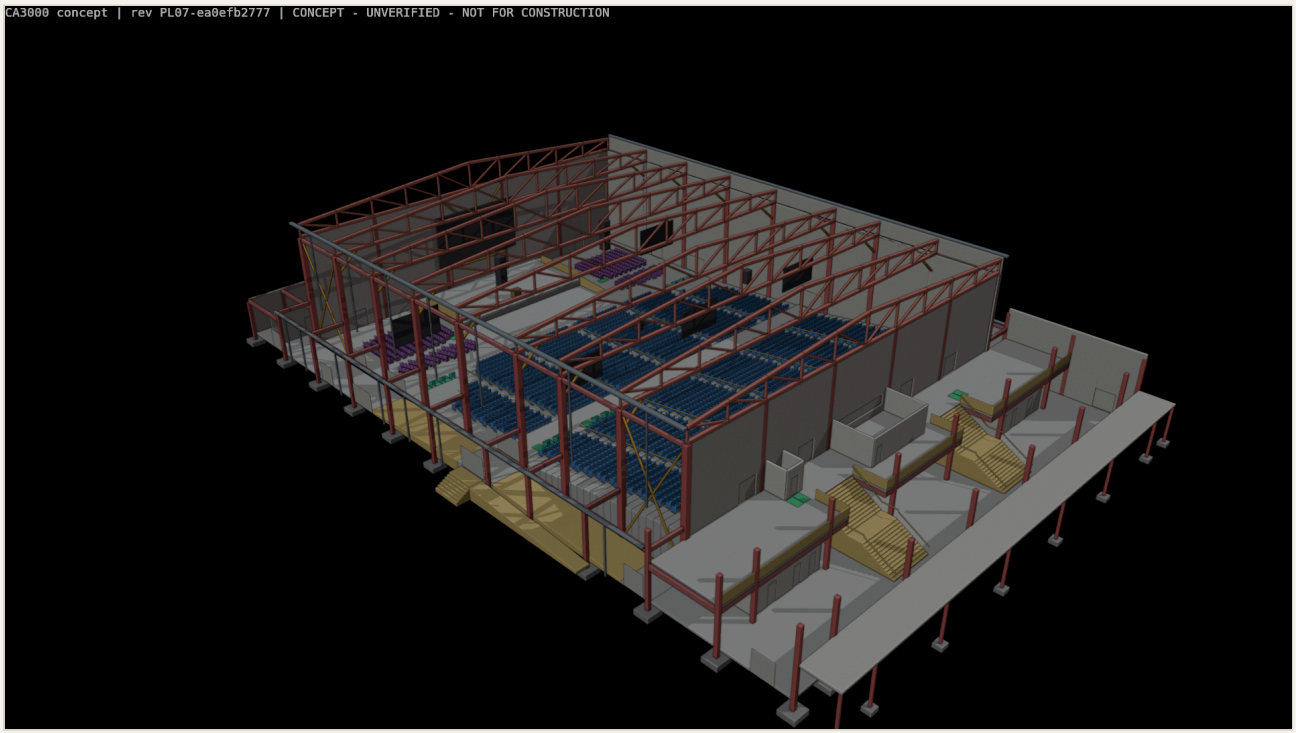
Concept coordination drawing generated from the Blender model. Not for construction. Dimensions unverified pending professional review.

Visualisations

Workbench renders of the saved model, stamped with the revision (renders from PL07-ea0efb2777; no renders were archived for PL08-590d273e35).



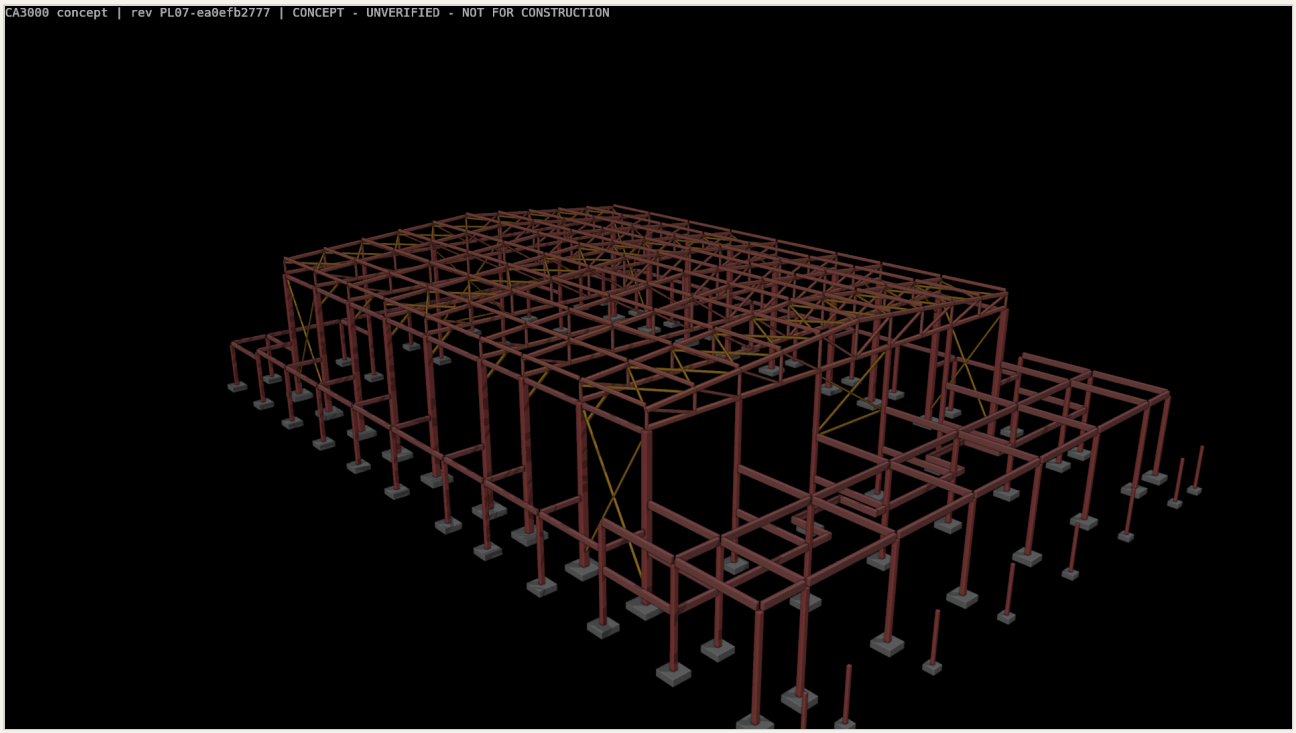
01 exterior aerial



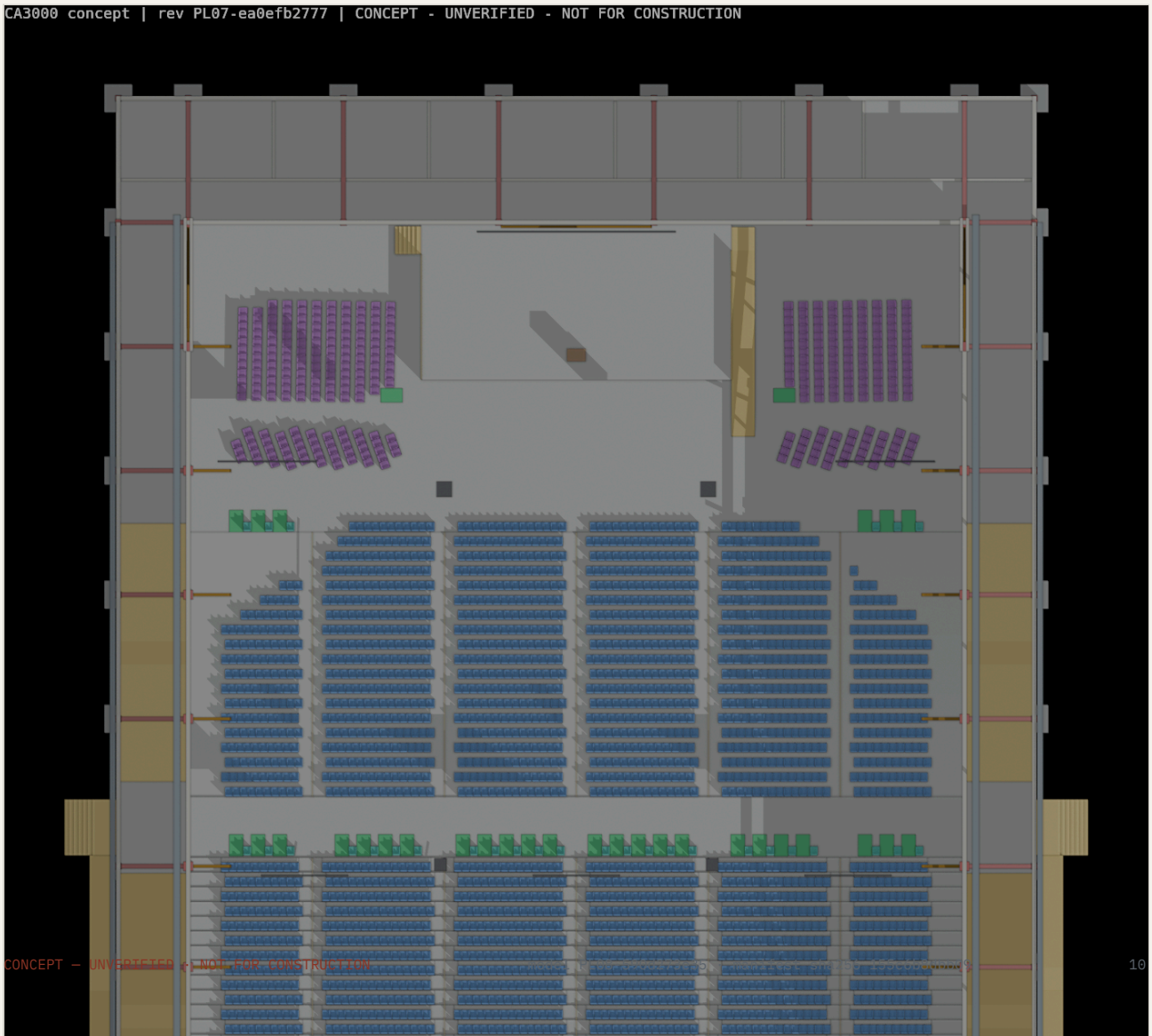
02 cutaway aerial



Visualisations (continued)



05 structure only



Seating schedule (rows)

reports/row_schedule.csv

row	section	seats	floor_elevation
1	front	49	0.0
2	front	53	0.025
3	front	56	0.055
4	front	57	0.09
5	front	62	0.13
6	front	67	0.175
7	front	72	0.225
8	front	76	0.275
9	front	76	0.335
10	front	76	0.395
11	front	76	0.46
12	front	76	0.525
13	front	76	0.595
14	front	76	0.665
15	front	76	0.74
16	front	76	0.815
17	front	76	0.895
18	front	76	0.975
19	front	76	1.06
20	rear	76	1.34
21	rear	76	1.48
22	rear	76	1.62
23	rear	76	1.76
24	rear	76	1.9
25	rear	76	2.04
26	rear	76	2.18
27	rear	76	2.32
28	rear	76	2.46
29	rear	76	2.6
30	rear	76	2.74
31	rear	76	2.88
32	rear	76	3.02
33	rear	76	3.16
34	rear	76	3.3
35	rear	76	3.44
36	rear	76	3.58
37	rear	76	3.72
38	rear	76	3.86
39	rear	76	4.0
40	rear	76	4.14

Egress and accessibility

reports/exit_schedule.csv, geometry_checks.json (placeholder factors, not code values)

exit_id	wall	clear_width_m	load_queue_balanced
DR-HALL-W-FRONT	W	3.36	435
DR-HALL-W-WING	W	2.24	272
DR-HALL-W-CROSS	W	3.36	473
DR-HALL-E-FRONT	E	3.36	437
DR-HALL-E-WING	E	2.24	272
DR-HALL-E-CROSS	E	3.36	474
DR-HALL-REAR-A1	REAR	2.24	265
DR-HALL-REAR-A2	REAR	2.24	308
DR-HALL-REAR-A4	REAR	2.24	307
DR-HALL-REAR-A5	REAR	2.24	265

Egress metric (own engine)	Value
occupants_assigned_in_hall	3060
platform_choir_added	448
total_hall_exit_width	26.88
total_required_width_placeholder	17.54
rear_load_to_upper_foyer	1145
upper_foyer_stair_width	10.8
upper_foyer_stair_required_placeholder	8.59
max_travel_hall	50.87
max_travel_hall_to_nearest_exit	39.73
count_nearest_travel_over_flag	330
max_travel_total	61.62
boh_staff	30
building_total_occupants	3538
count_travel_hall_over_flag	490
count_travel_total_over_flag	4

Risks and open issues

- Plan travel to the NEAREST hall exit reaches 39.7 m (330 occupants beyond the 30 m placeholder flag); to the queue-balanced exit 50.9 m. This is because the hall is 50 m wide and exits are on the side and rear walls. The 30 m / 60 m flags are placeholders, not local limits. Mitigations to be tested by the fire engineer: additional exits through the platform wings / stage rear, vomitories under the rear rake, sprinklers or smoke control as the local code allows, or a narrower/deeper plan.
- `EGR-01N` / exit_schedule.csv column `placeholder_check_if_nearest` show which doors would be overloaded if everyone used the nearest exit (iteration 2 widened the cross-aisle exits to 3.6 m by moving grid 6 to y=38.5); the queue-balanced assignment is a heuristic and must be replaced by a code-based egress analysis.
- Restriction on use: until OI-01/OI-05 are closed, this layout must not be used to fix occupant capacity, exit provision or fire strategy.

Independent review (iter-01)

comparisons/reviews/iter-01/A-*.json — QA scores, not certification.

Discipline	Score /10
accessibility	7.0
acoustics_av	5.5
aesthetics	4.0
architecture	6.0
church_functionality	5.5
constructability_maintenance	6.0
cost_drivers	6.5
life_safety	5.5
structural_engineering	7.0

Severity	Discipline	Finding
MAJOR	church_functionality	The 41-row end-stage hall puts a large part of the congregation far from the preacher: 1,261 seats are more than 40 m and 837 more than 45 m from the pulpit, so the rear half depends on IMAG screens to see the preacher's face. Pushing the platform back against
MAJOR	church_functionality	Straight, parallel rows in a 49.4 m wide hall make side seats turn their heads a long way: 670 seats turn more than 30 deg (max 52.5 deg) and 156 sit more than 45 deg off the pulpit. A fanned bowl was never studied.
MAJOR	architecture	Arrival and departure rely on the upper foyer. Rows 20-41 (about 1,600 people) enter and leave through rear exits at +4.42, then use two foyer stairs (10.8 m total) down to a ground foyer of only 484 m2 plus 225 m2 under the mezzanine. That is about 0.24 m2 pe
MAJOR	architecture	The sanitary allowance is far too small for 3,000+ occupants: two 34 m2 toilet blocks plus four 16 m2 accessible WCs, with no WCs on the upper foyer. This makes the foyer layout unreliable, because correctly sized toilets would take a lot of foyer area.
MAJOR	aesthetics	The exterior reads as a windowless industrial shed: an 88 m long gabled box with 9.6 m2 of glazing in total. The entrance is a low flat-roofed foyer block with a plain columned canopy, giving no civic or ecclesiastical presence, no daylight to the hall, and no
MAJOR	life_safety	The two upper-foyer stairs that carry the rear 1,145 occupants down from +4.42 m are single straight flights of 26 risers with no intermediate landing, open to the foyer (not enclosed or protected).
MAJOR	life_safety	Travel distances exceed the placeholder flags. They are honestly logged, but they remain the main weakness of a 50 m wide end-stage hall: 330 occupants are over 30 m from the nearest exit (max 39.7 m), 490 are over 30 m with queue balancing (max 50.9 m), and t
MAJOR	life_safety	If occupants take the shortest route to an external door, the two mid cross-aisle exits are overloaded (817 and 843 occupants against 3.36 m clear width). The queue-balanced assignment that passes is a heuristic, and occupants naturally favour these exits beca
MAJOR	acoustics_av	The hall volume is very large for amplified worship (49,768 m3, or 16.6 m3 per seat), and the hall is a box whose rear wall runs parallel to the platform about 48 m away. Without heavy absorption the reverberation will be long, and the rear wall will send a la
MAJOR	constructability_maintenance	Purlins are modelled only at truss panel points, i.e. at 5.0 m centres along the slope (5.02 m top-chord panels), spanning 8-9.5 m between trusses. Typical trapezoidal metal roof sheeting does not span ~5 m between purlins, so secondary purlins would be needed
MAJOR	constructability_maintenance	Hall eaves gutters sit at about +15.9 to +16.1 m and the 14 hall downpipes discharge onto the ambulatory roofs (cascade). The smaller ambulatory gutters (0.35 x 0.20 placeholder) then take hall plus ambulatory run-off above the ambulatories, which are also esc
MINOR	church_functionality	The 117 m2 floor in front of the platform is still marked 'unallocated'. As an altar-call / ministry floor it is a strength, but it is undeclared and leaves a 5.85 m dead gap between the preacher and the front row.
MINOR	church_functionality	The choir in the side wings (386 chairs) faces across the platform in straight rows. It sits at floor level below a 1.00 m platform, so for most of the congregation it is seen side-on and partly hidden. Choir-to-conductor and choir-to-band sightlines are not c
MINOR	aesthetics	The interior is dominated by exposed trusses, a blank rear wall and six hanging screens (three delay screens mid-hall). There is no focal architectural treatment behind the platform.
MINOR	architecture	No renders exist for the frozen baseline PL08. The latest renders are PL07, so the PL08 stair, ramp and choir-

Revision history

Option A (end stage) - revision history

Every revision ID is `<tag><nn>-<first 10 hex of SHA-256 over the source files>`. Archived outputs are in `archive/<rev>/`. Decisions are recorded in `../../DESIGN_BRIEF.md` (Client design decisions).

Revision	Date	Kind	Change	Sources archived
IT01-45b0e08c36	2026-11-01	review loop 1	first build	outputs only
IT02-325e67e273	2026-11-01	review loop 2	repairs	outputs only
IT03-1e06f520e1	2026-11-01	review loop 3	repairs; 8.79/10 NOT_ACCEPTED (see FINAL_REPORT.md)	outputs only
PL01-7cafe48b2e	2026-11-01	post-loop	side choir facing inward	outputs only
PL02-18dac43917	2026-11-01	post-loop	review response: trimmed off-axis seats, simpler foyer (two central stairs)	**yes - `archive/pl-02/source/`, reproduced**
PL03-3e57114669	2026-11-01	post-loop	choir risers removed	outputs only
PL04-7e854e086c	2026-11-01	post-loop	144 choir chairs in the wings	outputs only
PL05-cba999fc37	2026-11-01	post-loop	platform 20 x 10 m	outputs only
PL06-966bde8984	2026-11-02	post-loop	choir in four zones	outputs only
PL07-ea0efb2777	2026-11-02	post-loop	platform to rear wall, +1.00 m	outputs only
PL08-590d273e35	2026-11-02	post-loop	stair/ramp moves, straight choir rows (386 chairs) - **Option A baseline**	**yes - `archive/pl-08/source/`, reproduced**

`proposal/legacy-PL06-21c8b6d8b4/` holds the proposal data and drawings that were built (outside this pipeline) from a model labelled `PL06-21c8b6d8b4`. That revision ID matches no archived build; treat it as superseded and unverified.